Angled launches

antocciphysics | TEACHER KEY | CPA

Learning target: Use components to analyze an angled projectile.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 8; 3 course days later.*
10 m/s² downward. Timing resolution, sensor fit or release timing can contribute uncertainty.

02. Explain a worked example (worked)
Horizontal component 4 m/s throughout; vertical +10, 0,-10 m/s at the three marked times.

03. Look and explain (model)
Vertical velocity zero; horizontal remains; acceleration 10 m/s² down.

04. Build the solution (practice)
1 s; 2 s; 12 m.

05. Use the idea (reason)
Vertical motion is symmetric about top at equal height. A different landing height changes remaining fall time.

06. Choose a method (mixed) *Review: Unit 1, Day 20; 13 course days later.*
$a=9$ m/s² right; $v_f=54$ m/s right. Constant net force and mass throughout.

07. Exit check (exit)
1.5 s and 3 s.

08. STRETCH: Apply it another way (stretch)
Time and height unchanged; range doubles to 16 m under the same ideal assumptions.

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01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 8; 3 course days later.*
$g=\Delta v/\Delta t=2.94/0.30=9.8$ m/s². Several points reveal a trend and reduce sensitivity to one noisy observation.

02. Explain a worked example (worked)
Horizontal component 4 m/s throughout; vertical +10, 0,-10 m/s at the three marked times.

03. Represent and interpret (model)
Vertical velocity zero; horizontal remains; acceleration 10 m/s² down.

04. Solve and justify (practice)
v_x 17.32, v_y 10 m/s; time 2 s; range 34.6 m; max height 5 m.

05. Analyze the reasoning (reason)
Vertical motion is symmetric about top at equal height. A different landing height changes remaining fall time.

06. Choose a method (mixed) *Review: Unit 1, Day 20; 13 course days later.*
x interval [-77, -63] km; y interval [81, 99] km. Simplified geometry, gravity, Earth motion or missing observations limit the model. Bounds are not probabilities.

07. Exit check (exit)
Both 34.6 m; 60° flight time 3.46 s versus 2 s at 30°. Complementary angles same ideal range.

08. STRETCH: Apply it another way (stretch)
$\sin(2\theta)=0.5$ gives 15° and 75°. Times 4 $\sin\theta\approx1.04$ and 3.86 s; same range, different arcs.